

Report on The Cost of Living (US)

By Devin Frost

In recent years, the US economy has been hit hard by multiple economic crises, from global pandemics to supply chain shocks to wars, and naturally, the expenditure of resources to combat these crises has negatively impacted the cost of living in the US. This report takes a holistic look into the cost of living in the United States, using data from the Economic Policy Institute (EPI). In order to understand the state of the US economy, we first need to measure the cost of living, then normalize it in accordance to median incomes in order to gauge regional affordability.

To prepare my data for this analysis, I dropped rows with null values to simplify the analysis process. It only meant losing 10 records, which should not meaningfully offset the value of this analysis. I also synthesized numerous measures using the dataset. The dataset gives a simple 'family_member_count' field, which isn't provided as a numerical value. I parse the values into the 'parents' and 'children' fields. Further, I sum these up into a 'family_size' value. In order to measure the affordability of a region, I take total expenditures and divide it by median income, to normalize the values.

The family unit is not complete without children, and in the modern, fast-paced world, childcare is a necessity. Figure 1 is a map of childcare costs by state. Childcare costs have considerable variation across the states. Figure 2 is a bar chart of the top 12 most expensive counties in which to raise a child. It includes counties from New York and the District of Columbia.

Housing. We all need it, whether we are high up in the mountains of Appalachia or in the Great Plains - to simply survive the elements. Figure 4 is a histogram that shows the typical annual cost of housing per state. Figure 5 shows housing prices by state on a map, and as you can see, housing prices are generally highest near the coasts, and lowest in the Midwest and Southeast. Hawaii, being a small slice of Heaven on Earth, naturally has the highest housing prices, with a median of 61k / year. Country-wide it floats at a median of \$10,400 each year. Figure 6 is a scatterplot of housing prices versus taxes, and suggests that higher housing costs are positively correlated with higher taxes, which is likely due to property tax law.

A well-functioning economy depends on cheap transportation to move goods, services, and people to where the demand is. Figure 7 shows the distribution curve of transportation costs across America. As you can see, the graph is left skewed with a median of approximately \$13,800 a year, which, as shown by Figure 8 is due to the outlier Washington, D.C., where transportation is cheap. Figure 9 is a regression plot of transportation cost versus total expenditures, and it shows that transportation costs are negatively correlated with total costs.

This may indicate that cheaper transportation leads to more economic growth and therefore higher paying jobs and cost of living

Our quality of life is impacted largely by our health, and consequently, regions with high healthcare costs are indicative of regions that struggle to meet a standard of living. Figure 10 is a histogram of the distribution of healthcare expenditures accross the country. It is somewhat right-skewed, but is largely normally distributed. Figure 11 shows that West Virginia is by far the most expensive place to pay for healthcare, with a median price of \$37,000 annually. It is reflective of poor policy decisions in the region.

Unfortunately, in order to organize society, any state needs a government, and any government needs tax dollars to power programs that power public infrastructure and social programs. Figure 12 is a bar chart of taxes by state, and Figure 13 is a map thereof. As shown by the map, Oregan and Washington, DC have some of the highest taxes in the country. Figure 14 is a bar chart of the top 12 highest-tax counties in the country. Counties from New York and Washington, D.C have the highest taxes of any region in the United States

In order to pay for all of this, people naturally need sturdy, competitive incomes. Figure 15 is a boxplot of median household incomes across the counties of the 50 United States, plus the District of Columbia, and it shows that incomes across the state are highly variable, ranging from \$27,000 in Georgia, up to \$177k in Virginia. Virginia was by far the most riddled with inter-county income inequality. Deleware had the least variation. Figure 16 better conveys this information geographically. Figure 17 is a histogram of median family incomes by state and it is evident that the median wage in the US is approximately \$70,000 a year.

Figure 19 is a correlation matrix between each of the aforementioned factors and concludes that family size is highly correlated with the cost of food and healthcare, that total costs are highly correlated with the cost of food and the cost of extraneus necessities, and, interestingly, housing costs are correlated with extraneuss necessity expenditures. Figure 18 shows that total expenditures are positively correlated with higher taxes. A small table (Figure 20) is provided to illustrate the data that were provided by the Economic Policy Institute (EPI), including the aforementioned feilds `parents`, `children`, `family\_size`, and `is\_foodcost\_outlier`.

Childcare

Figure 1



Figure 2

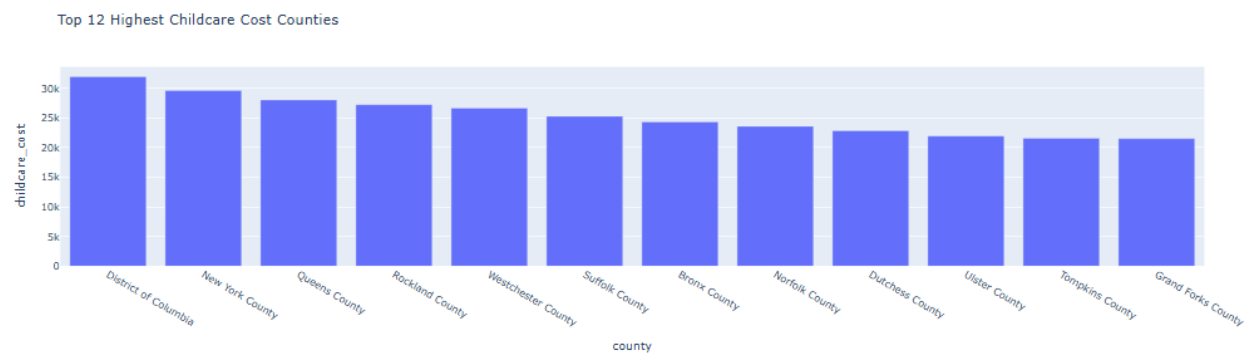
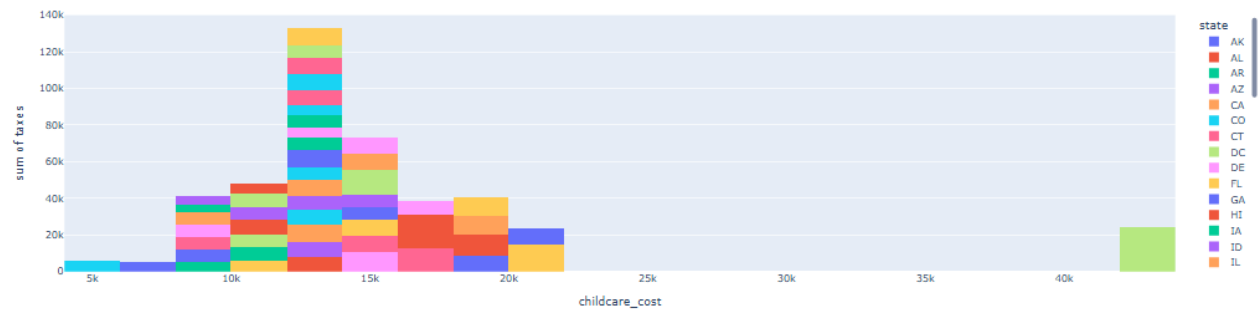


Figure 3



Housing

Figure 4

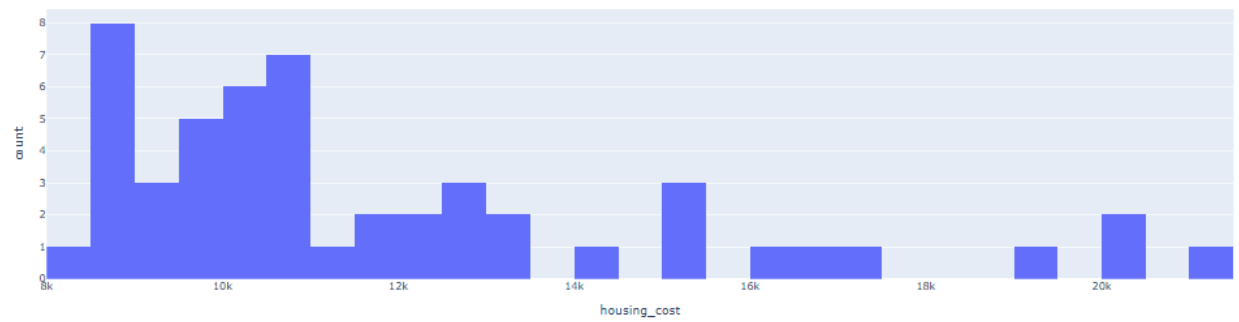


Figure 5

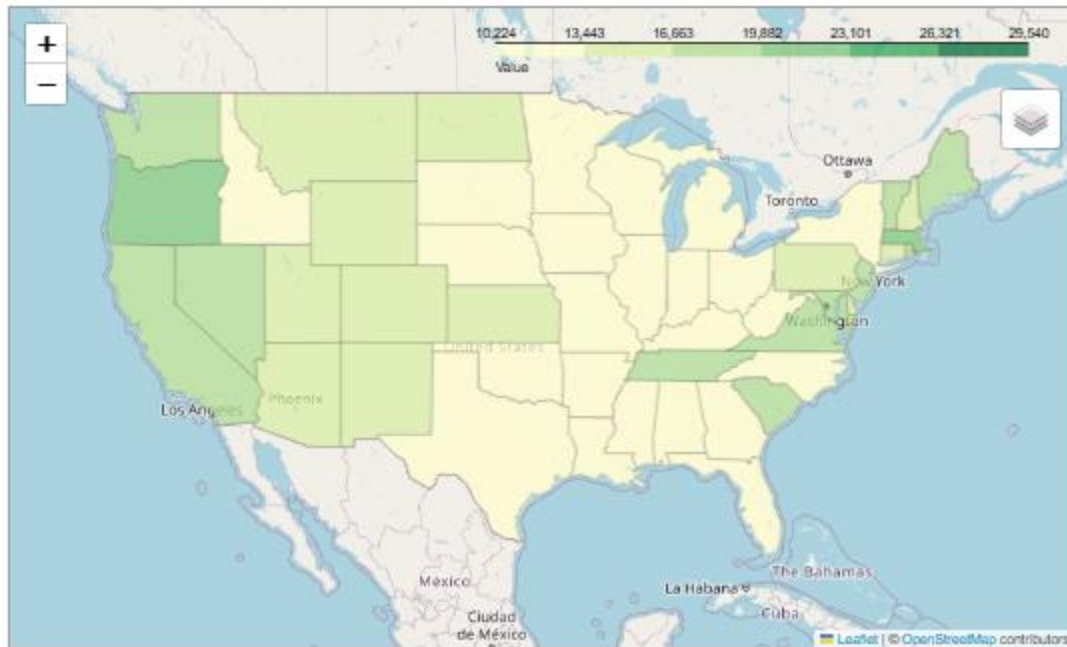
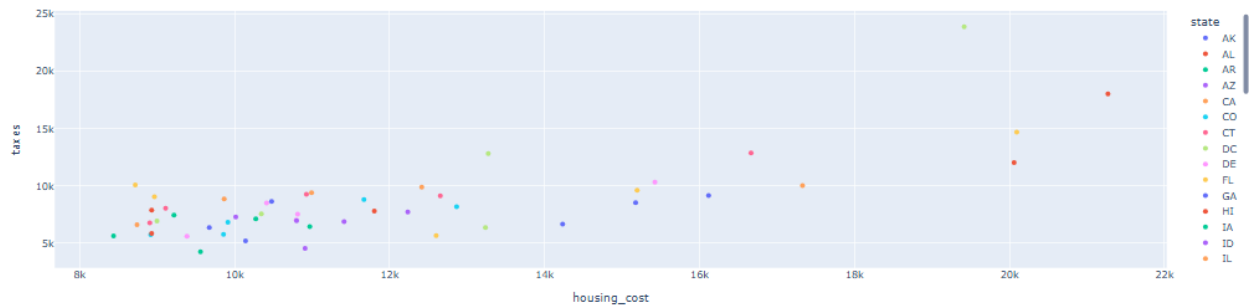


Figure 6



Transportation

Figure 7

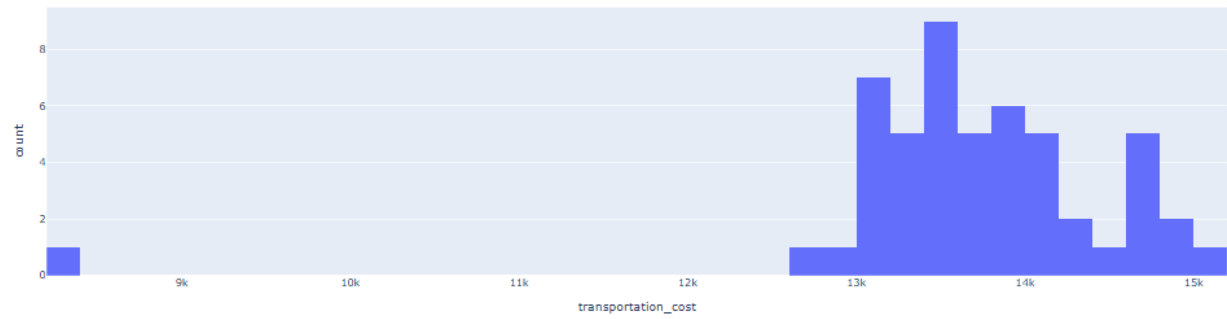


Figure 8

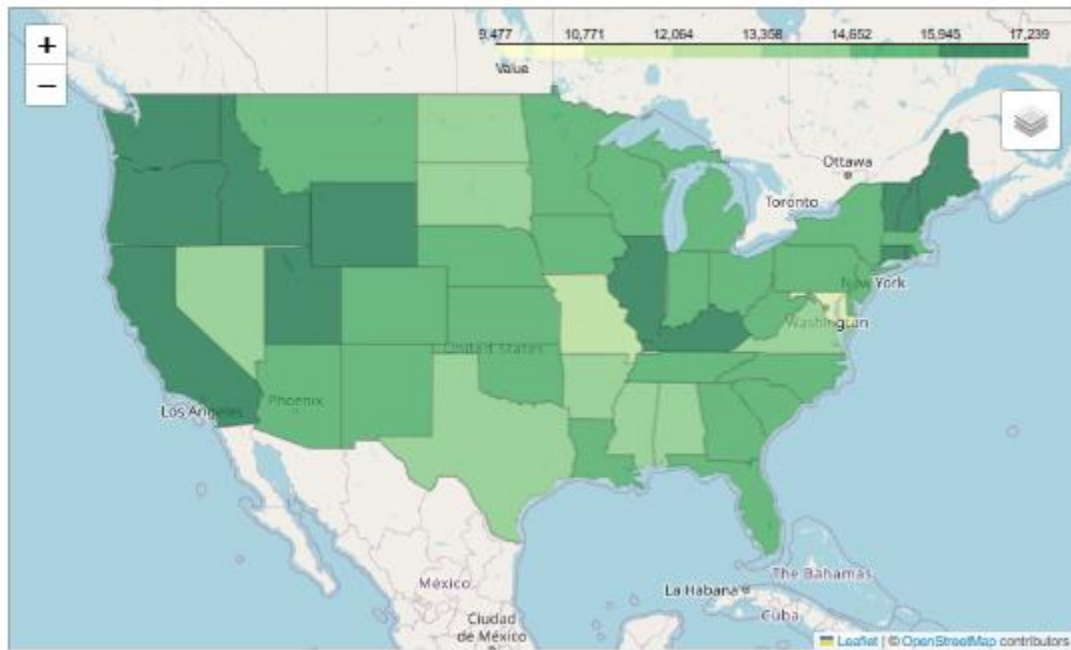
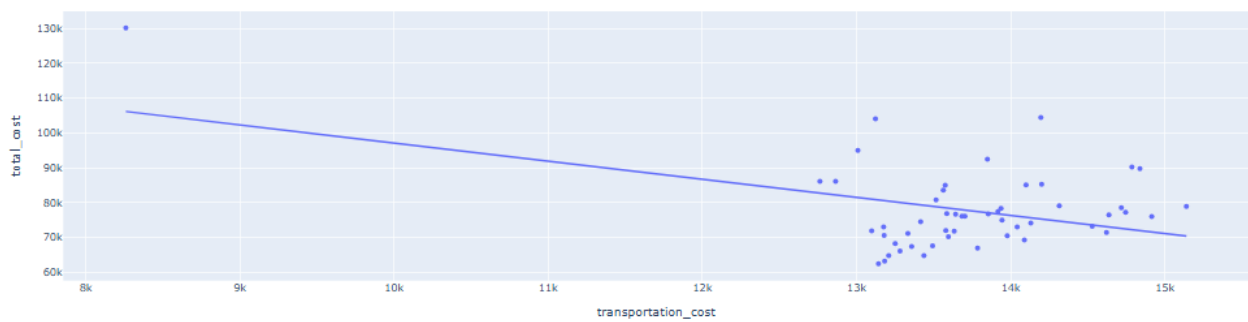


Figure 9



Healthcare

Figure 10

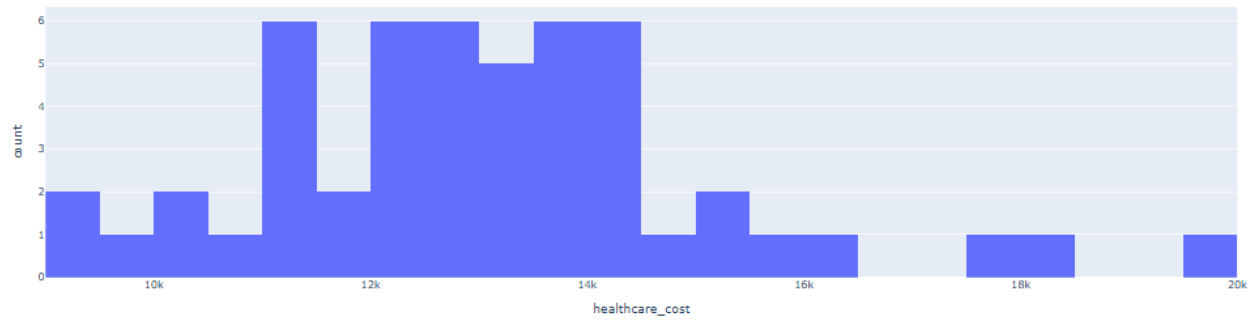
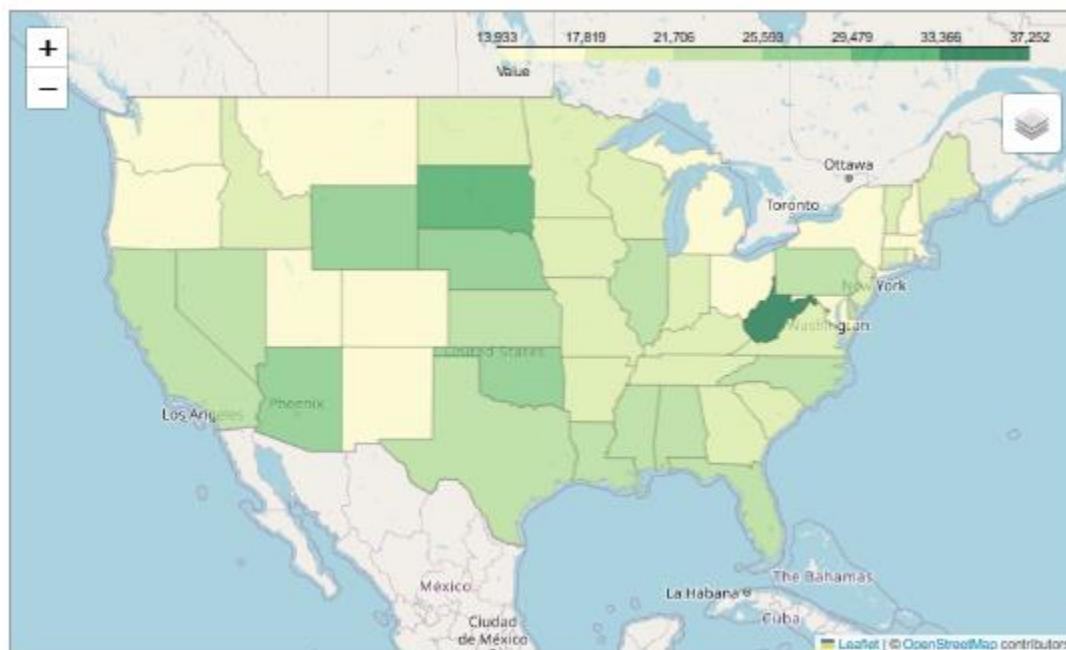


Figure 11



Taxes

Figure 12

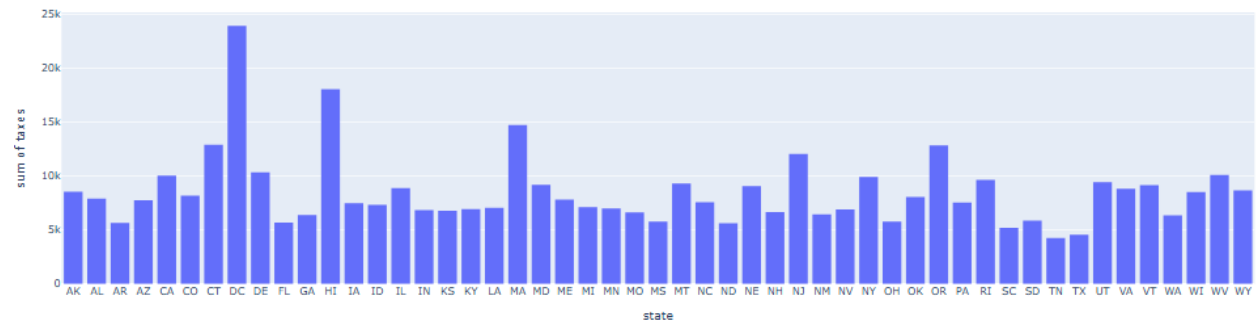


Figure 13

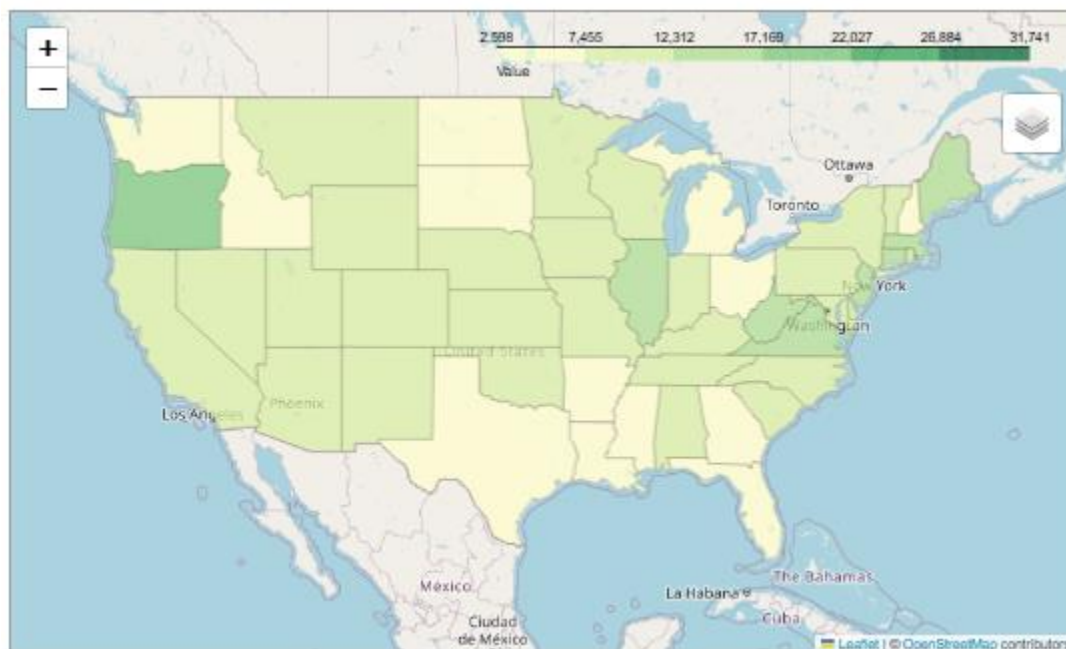


Figure 14



Income

Figure 15

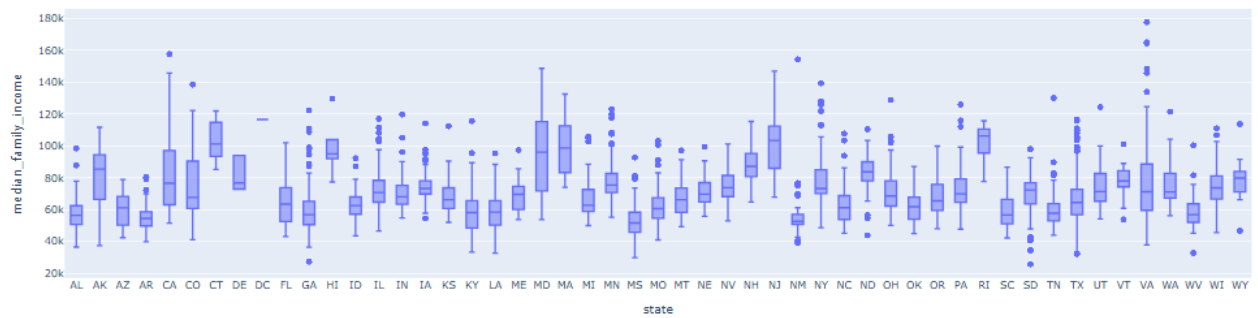


Figure 16

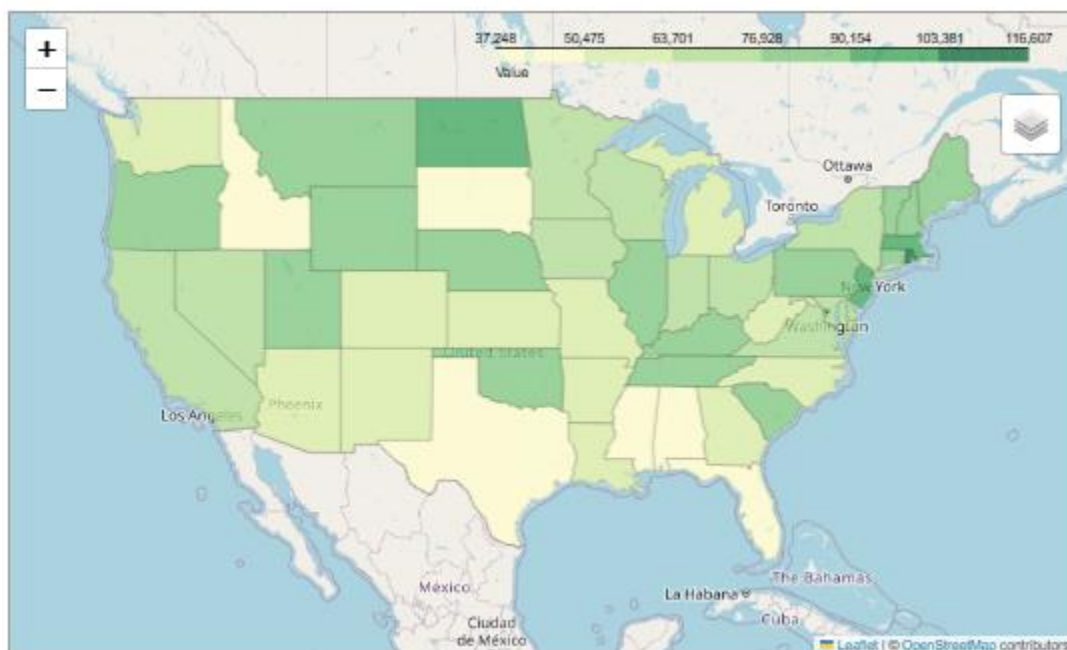
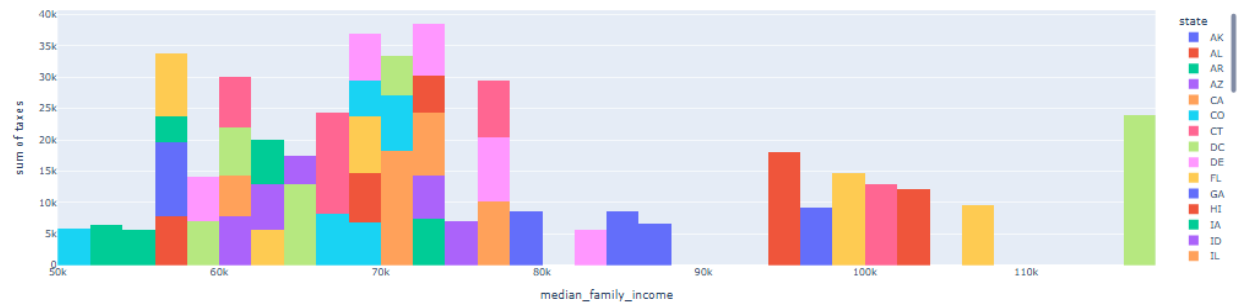


Figure 17



Summary

Figure 18

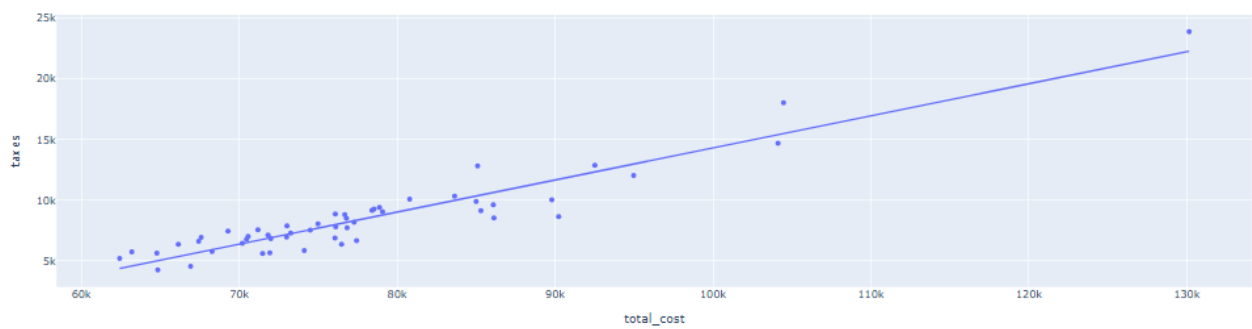


Figure 19

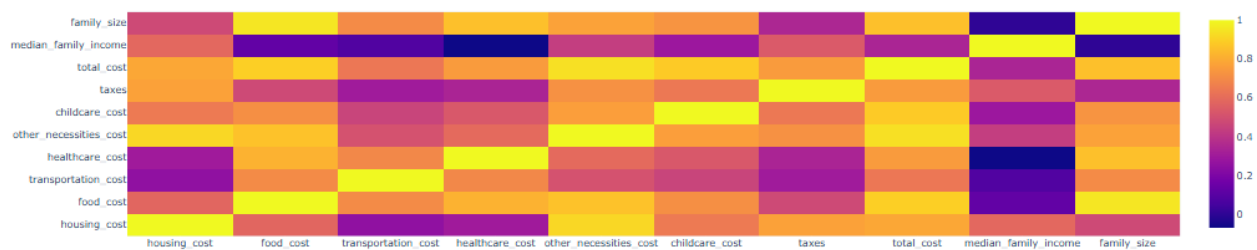


Figure 20

state	isMetro	areaname	county	family_m	housing_c	food_cost	transporta	healthcare	other_nec	childcare	taxes	total_cost	median_f	state_full	parents	children	family_size	is_foodcost
AL	true	Montgomery AL MSA	Autauga County	1p0c	8505.728	3454.917	10829.16	5737.479	4333.813	0	6392.945	39254.05	73010.41	Alabama	1	0	1	false
AL	true	Montgomery AL MSA	Autauga County	1p1c	12067.50	5091.707	11588.19	8659.556	6217.458	6147.829	7422.078	57194.32	73010.41	Alabama	1	1	2	false
AL	true	Montgomery AL MSA	Autauga County	1p2c	12067.50	7460.203	12361.77	11581.63	7075.658	15824.69	9769.562	76141.03	73010.41	Alabama	1	2	3	false
AL	true	Montgomery AL MSA	Autauga County	1p3c	15257.15	9952.239	13452.18	14503.70	9134.356	18802.18	13101.70	94203.53	73010.41	Alabama	1	3	4	false
AL	true	Montgomery AL MSA	Autauga County	1p4c	15257.15	12182.21	13744.59	17425.78	9942.363	18802.18	13469.21	100823.5	73010.41	Alabama	1	4	5	false